Unit 1 Classwork — Real Numbers — Answer Key

Converting · Classification · Squares · Estimating Roots · Ordering

A. Converting Fractions ↔ Decimals

Problem 1 Score 3

Convert $\frac{3}{8}$ to a decimal.

Answer: 0.375

Problem 2 Score 3

Convert 0.45 to a fraction in simplest form.

Answer: $\frac{9}{20}$

Problem 3 Score 4

Convert $\frac{5}{6}$ to a decimal (round to the thousandths, use a repeating bar notation if needed).

A. $0.8\overline{3}$
B. 0.83
C. $0.\overline{83}$
D. 1.2

Answer: A

Problem 4 Score 4

Convert $0.\overline{6}$ (repeating) to a fraction.

Answer: $\frac{2}{3}$

B. Classifying Numbers

Problem 1 Score 3

Is √16 rational or irrational?

Answer: rational

Problem 2 Score 3

Is √7 rational or irrational?

Answer: irrational

Problem 3 Score 3

Is π rational or irrational?

Answer: irrational

Problem 4 Score 4

Is 0.454545... (repeating) rational or irrational?

Answer: rational

Problem 5 Score 4

Is −5 a whole number, an integer, or both?

Answer: integer

Problem 6 Score 5

Is $\frac{3\pi}{4}$ rational or irrational? Pick the best reason.

A. Rational. 3 and 4 are integers.

B. Irrational. Every fraction is irrational.

C. Irrational. π is irrational, so 3π/4 is not a ratio of two integers.

D. Rational. Any number written as a fraction is rational.

Answer: C

C. Perfect Squares

Problem 1 Score 3

What is 12^2?

Answer: 144

Problem 2 Score 3

What is $\sqrt{225}$?

Answer: 15

D. Estimating Square Roots

Problem 1 Score 3

Between which two consecutive integers does $\sqrt{50}$ fall?

Answer: 7 and 8

Problem 2 Score 4

Estimate $\sqrt{50}$ to the nearest tenth.

Answer: 7.1

Problem 3 Score 5

Kai estimates $\sqrt{40} \approx 6.3$. Which check shows the estimate is good?

A. $6.3 \times 2 = 12.6$, which is not close to 40.

B. $40 \div 2 = 20$, so 6.3 is too low.

C. $6^2 = 36$ and $7^2 = 49$, so any number from 6 to 7 is a good estimate.

D. $6.3^2 = 39.69$, which is very close to 40.

Answer: D

E. Ordering Rational & Irrational Numbers

Problem 1 Score 4

Which is greater: $\sqrt{2}$ or $\frac{3}{2}$?

Answer: $\frac{3}{2}$

Problem 2 Score 5

Order from least to greatest: $\sqrt{10}$, 3, $\frac{10}{3}$

Answer: 3, $\sqrt{10}$, $\frac{10}{3}$

Problem 3 Score 5

Which is greater: π or $\sqrt{10}$?

A. π

B. $\sqrt{10}$

C. They are equal

Answer: B

Problem 4 Score 5

Order from least to greatest: $\sqrt{30}$, $\sqrt[3]{100}$, 5.3

Answer: $\sqrt[3]{100}$, 5.3, $\sqrt{30}$